

A type A_2 counterexample to conjecture 00000000564

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Abstract

Conjecture 00000000564 asserts that the specialization $F_i(1)$ of an F -polynomial divides the corresponding generalized Catalan number. In the principal-coefficient cluster algebra of type A_2 , a single mutation produces $F_i = 1 + y_1$. Its value at $y_1 = y_2 = 1$ is 2, whereas the type A_2 generalized Catalan number is 5. Thus the asserted divisibility already fails in rank two.

1 Conventions and the precise instance

For a finite Coxeter group W of rank n , Coxeter number h , and exponents $e_1, \dots, e_n$, the generalized Catalan number is

$$\text{Cat}(W) = \prod_{j=1}^n \frac{h + e_j + 1}{e_j + 1}.$$

Here “generalized” refers to the extension to finite Coxeter types; there is no additional Fuss parameter in the conjecture. In type A_2 , $W \cong S_3$, $h = 3$, and $(e_1, e_2) = (1, 2)$. These are equivalently the standard type A_2 degrees $(2, 3)$. For a direct check of the Coxeter data, the simple reflections in the simple-root basis and their product are

$$s_1 = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, \quad C = s_1 s_2 = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

The matrix C has order 3 and characteristic polynomial $t^2 + t + 1$, whose roots are the two nontrivial cube roots of unity. This gives $h = 3$ and exponents 1, 2; these matrix and coefficient computations are also checked in Lean. Consequently

$$\text{Cat}(A_2) = \frac{3+1+1}{1+1} \frac{3+2+1}{2+1} = \frac{5}{2} \cdot \frac{6}{3} = 5. \tag{1}$$

We use the standard definition of F -polynomials for a cluster algebra with principal coefficients: for a cluster variable $x_{i;t}$, its F -polynomial is obtained by setting all initial cluster variables $x_1, \dots, x_n$ equal to 1, while retaining the coefficient variables $y_1, \dots, y_n$. Thus $F_i(1)$ means specialization of all coefficient variables to 1.

2 One mutation gives the counterexample

Consider the rank-two initial exchange matrix and its principal extension

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \tilde{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Its Cartan companion is $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, the Cartan matrix of type A_2 . The frozen variables associated with the last two rows are y_1, y_2 . For an initial seed with principal coefficients, the exchange relation at index k is

$$x_k x'_k = y_k \prod_{j=1}^n x_j^{[b_{jk}]_+} + \prod_{j=1}^n x_j^{[-b_{jk}]_+}, \quad [a]_+ = \max(a, 0). \quad (2)$$

This is also the normalized principal-coefficient relation because $y_k \oplus 1 = 1$ in the initial tropical semifield. Reading the first column of $\tilde{B}$ and mutating at $k = 1$ yields

$$x_1 x'_1 = y_1 + x_2, \quad x'_1 = \frac{y_1 + x_2}{x_1}.$$

Specializing $x_1 = x_2 = 1$ gives the actual F -polynomial of this cluster variable:

$$F_{1;\mu_1(t_0)}(y_1, y_2) = y_1 + 1, \quad F_{1;\mu_1(t_0)}(1, 1) = 2. \quad (3)$$

Only this single initial mutation is needed; there is no extrapolation from numerical experiments.

Proposition 1. *Conjecture 00000000564 is false for the standard finite-type generalized Catalan number.*

Proof. The seed above has finite type A_2 . Equations (1) and (3) give an F -polynomial whose specialization equals 2 and whose corresponding generalized Catalan number equals 5. An integer multiple of 2 is even, while 5 is odd; hence 2 does not divide 5. This contradicts the asserted universal divisibility. $\square$

3 Formal verification

The accompanying dependency-free Lean 4 project makes the initial principal-coefficient mutation formula explicit, computes its specialization for the displayed rank-two seed, and computes the Catalan product using exact rational arithmetic. It then proves nondivisibility and refutes the divisibility statement for this particular type and mutation. A universal conjecture about all finite types and their F -polynomials necessarily includes this instance. The standard identification of the displayed Cartan matrix and Coxeter data with type A_2 is stated explicitly above; the formalization uses that concrete standard data, rather than a library of general cluster algebras or finite Coxeter groups. The submission README gives the build commands, theorem names, and their correspondence with the equations in this report.

Definition reference

The principal-coefficient definition of F -polynomials and the exchange relation used here are standard. See S. Fomin and A. Zelevinsky, *Cluster algebras IV: Coefficients*, available at [arXiv:math/0602259](https://arxiv.org/abs/math/0602259). All computations needed for the counterexample have been included.